

Design Areas – WhisperCrush

Product

WhisperCrush is a mobile application designed for anonymous mutual interest detection among social media users. It allows individuals to discreetly express romantic or personal interest in others without direct messaging or public exposure. Employing advanced privacy-preserving matching algorithms, biometric authentication, and random profile mixing to prevent timing or profiling attacks, WhisperCrush ensures that identities are only revealed when both parties indicate mutual interest.

Similar Existing Services

Several apps exist that offer anonymous crush or interest matching, including:

- Facebook’s Secret Crush feature – This product was launched in 2019, but was limited to the Facebook Dating ecosystem. It allowed users to select up to nine friends, creating matches only when interest was mutual, but struggled with privacy concerns and limited adoption.
- Traditional dating applications (e.g., Tinder, Bumble, Hinge) – These platforms dominate the dating market, though they primarily connect strangers rather than existing social connections, leaving a significant gap in the market for friend-to-romance transitions.
- Anonymous messaging platforms (e.g., Whisper, Session, and SayHi Chat) – These applications support anonymous messaging and confessions but not mutual interest matching.

Despite these, WhisperCrush differentiates itself by bridging the current gap between matchmaking platforms and anonymity. Moreover, WhisperCrush would implement strong privacy-preserving algorithms, biometric authentication, support for real-time updates, and create unique emotional safety—all features which no currently existing single offering combines.

Stakeholders

- *Users/Customers:* Social media users seeking discreet mutual interest validation. They value privacy, anonymity, emotional safety, and ease of use. They are motivated by the desire for safe, stigma-free romantic interaction without direct confrontation or rejection.
- *Owners and Investors:* They are interested in profit, user growth, retention, and brand reputation. They are motivated by market demand in dating/interest apps with privacy niches. They value scalability, security compliance, and monetization.
- *Employees/Developers:* They are focused on building, maintaining, and improving secure, bug-free, performant apps. They are motivated by technical challenge, innovation, and user impact.

- *Regulators:* They are concerned with compliance to laws such as GDPR and CCPA and security standards. They are motivated to protect user data, prevent misuse, and enforce transparency.
- *Social Media Platforms (Indirect):* They may have interest in integrations or may see WhisperCrush as a complementary or competitive service. Their policies affect data access or API usage permissions.
- *Security Researchers and Privacy Advocates:* They are interested in the app's contribution to privacy-preserving technology and in its adherence to ethical standards.

User Journey

The user journey for the primary user of WhisperCrush involves:

1. Sign-up and authentication: User download app, completes biometric authentication for secure identity proof without compromising anonymity in matches.
2. Interest expression: User anonymously flags interest in one or more social media profiles by entering identifiers. The app uses encrypted secure matching and random profile mixing to conceal timing and any direct tracking.
3. Bot confidant interaction: User 'confides' interest or feelings privately to the bot as a safe place for vulnerable expression. Bot retains privacy and informs user only of mutual interest matches without revealing other details.
4. Mutual match notification: Upon mutual interest detection, both users receive real-time anonymous notification allowing them to initiate conversation or connect on chosen terms, preserving control and anonymity until readiness for disclosure.
5. Connection management: Users manage ongoing interactions, maintain privacy preferences, and control visibility or discoverability settings.

This journey creates the following value tensions between stakeholders:

- Privacy vs. Transparency: Users desire complete anonymity until mutual interest, but regulators and security standards require some level of identity verification to avoid abuse. Biometric authentication balances this.
- User safety vs. Misuse: While anonymity empowers users, it can be exploited for harassment. The app design must include moderation and abuse prevention aligned with stakeholder values.
- Real-Time Interaction vs. Privacy Protection: Real-time updates create tension with timing attack vulnerability. Random profile mixing and encryption techniques would mitigate this risk.